

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 203 — Past Questions & Materials
by Renance

Course: GET 203
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

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Paper 1 of 3 — GET 203 Engineering Graphics and Solid Modeling II

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
FIRST SEMESTER EXAMINATION 2024/2025 ACADEMIC SESSION

COURSE TITLE: ENGINEERING GRAPHICS AND SOLID MODELLING II
COURSE CODE: GET 203

TIME: 3 hours

UNITS: 2

TRUCTION: ANSWER THREE QUESTIONS ONLY.

1. Figure 1 shows details from a stationary engine. Viewing from the arrow position, draw the assembled parts of the knuckle joint as shown.

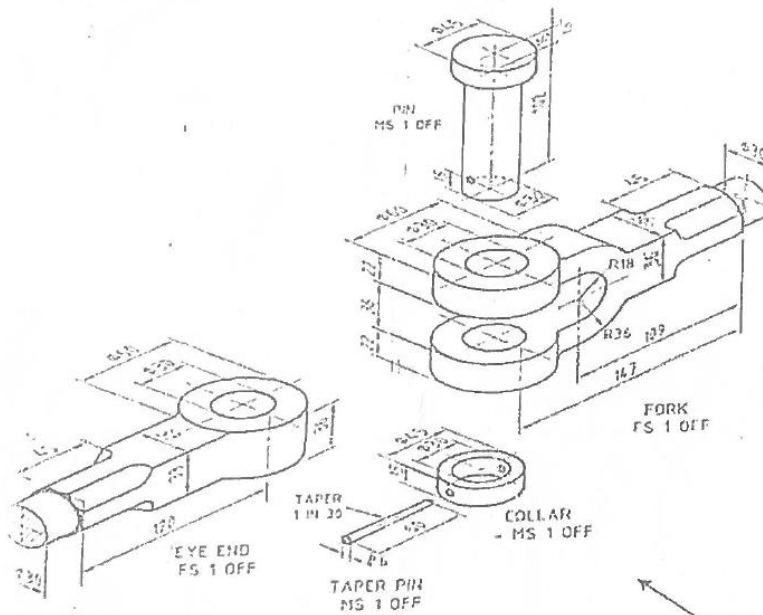
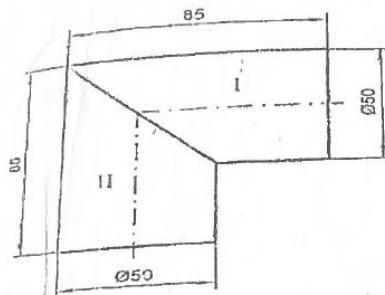


Figure 1: Knuckle Joint

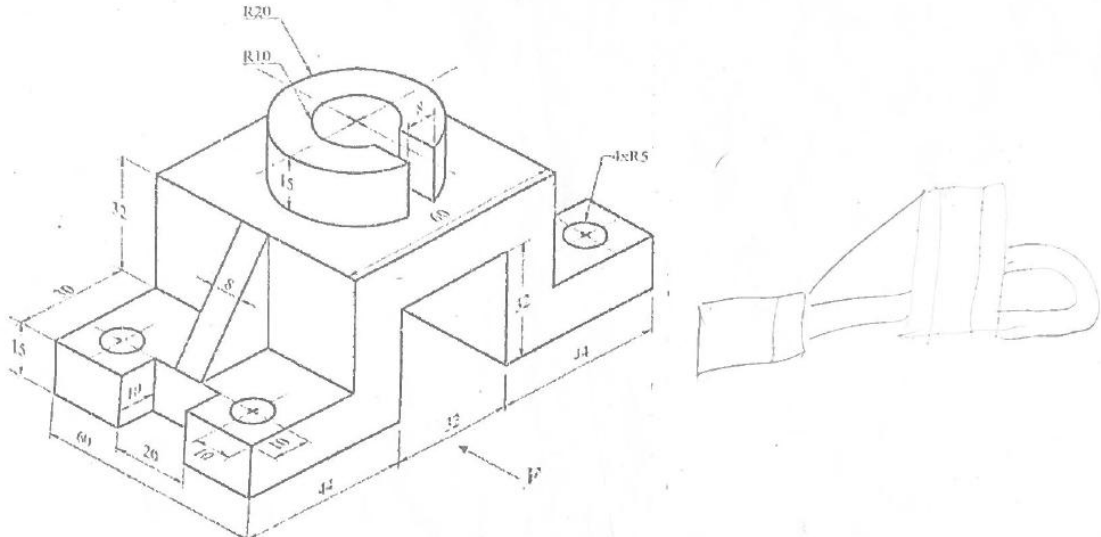
2. Draw the view and produce the surface development of a cylindrical elbow as shown in the figure below:



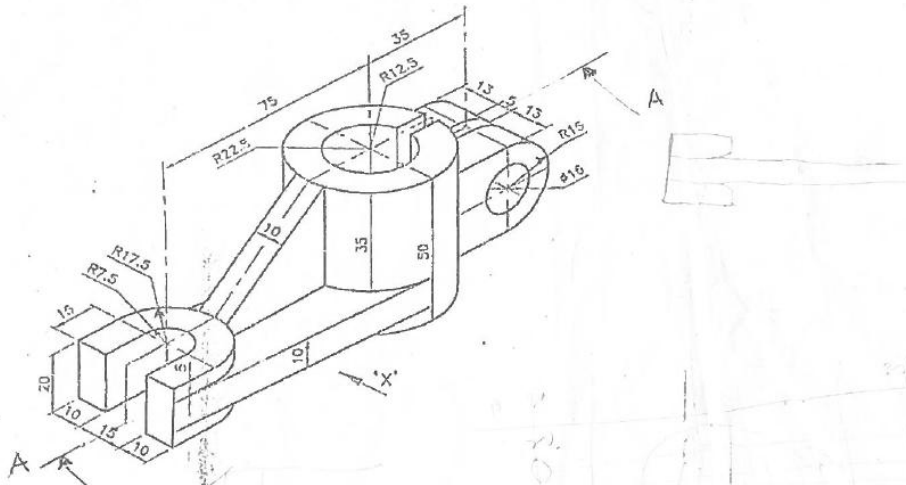
A section of a cylindrical metallic pipe of 75 mm length and diameter of 46 mm is penetrated by another cylindrical pipe of the same diameter at an angle of 60° . The center of the two cylinders coincide at 30

mm from the base of the section and the penetrating cylinder protrudes by 25 mm from the surface of the penetrated cylinder. Draw the front view and the plan and produce the curve of penetration on the surface of the elevation.

4. For the given solid as shown, draw the top and front views only in first angle projection.



5. The figure below shows the pictorial view of a machine component. Draw the sectional view AA as shown (ii) Also, draw the front view only viewing from point X



Paper 2 of 3 — GET 203 - 2024/2025 - First Semester - BASIC FLUID MECHANICS

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING
FIRST SEMESTER EXAMINATION 2024/2025 SESSION
COURSE CODE AND TITLE: GET 203 - BASIC FLUID MECHANICS
TIME ALLOTTED: TWO (2HRS) HOURS. INSTRUCTION: ANSWER ALL QUESTIONS

QUESTION 1 [17.5 Marks]

- The force of drag, F_D , on an air foil is due to flow of the foil field, the foil has a length, L . Find dimensionless number(s) using the Buckingham Pi theorem.
- A new design is proposed for an automobile. It is suggested that a 1:5 scale model study be conducted to assess the proposed design for a speed of 90 km/h. What speed should be selected for model study and what drag force would be expected on the prototype if a force of 80 N

QUESTION 2 [17.5 Marks]

- A heavy car plunges into a lake during an accident and lands at the bottom of the lake wheels as shown in Figure 2a. The door is 1.2 m height and 1m wide, and the top edge of the door is 8 m below the free surface of the water. Determine the hydrostatic force on the door location of the pressure center.

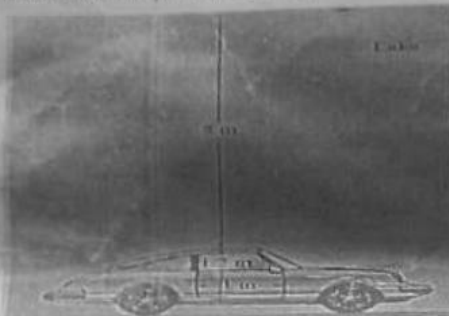


Figure 2a: The Heavy Car

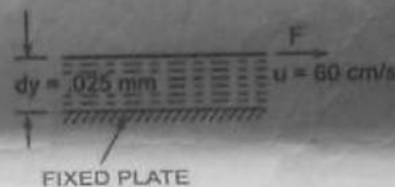


Figure 2b: Fixed Plate

- A plate 0.025 mm distant from a fixed plate moves at 60 cm/s and requires a force of 2N/ maintain this speed as depicted in Figure 2b. Determine the fluid viscosity between the plates.

QUESTION 3 [17.5 Marks]

- With the aid of a shear stress versus velocity gradient, compare and contrast ideal fluid, real fluid, Newtonian fluid, Non-Newtonian fluid, and ideal plastic fluid.
- A 15 cm diameter vertical cylinder rotates concentrically inside another cylinder with a diameter of 15.10 cm. Both cylinders are 25 cm high. The space between the cylinders is filled with fluid of unknown viscosity. If a torque of 12.0 Nm is required to rotate the inner cylinder at 1 determine the viscosity of the fluid.

QUESTION 4 [17.5 Marks]

- Calculate the dynamic viscosity of an oil used to lubricate a square plate of size 0.8 m x 0.8 m on an inclined plane with an angle of inclination as shown in Fig. 3a. The weight of the square plate is 300 N, and it slides down the inclined plane with a uniform velocity of 0.3 m/s. The thickness of the oil film is 1.5 mm.
- Define any five (5) of the following terms or explain engineering component briefly: (i) cavitation, (ii) capillary action, (iii) venturimeter, (iv) weirs, (v) orifices, (vi) dimensionless analysis, (vii) elbows, (viii) bends

Cavitation

Paper 3 of 3 — GET 203 ENGINEERING GRAPHICS AND SOLID MODELLING II 2024I2025

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY

FIRST SEMESTER EXAMINATION 2024/2025 ACADEMIC SESSION

JRSE TITLE: ENGINEERING GRAPHICS AND SOLID MODELLING II

TIME: 3 hours

JRSE CODE: GET 203

UNITS: 2

INSTRUCTION: ANSWER THREE QUESTIONS ONLY.

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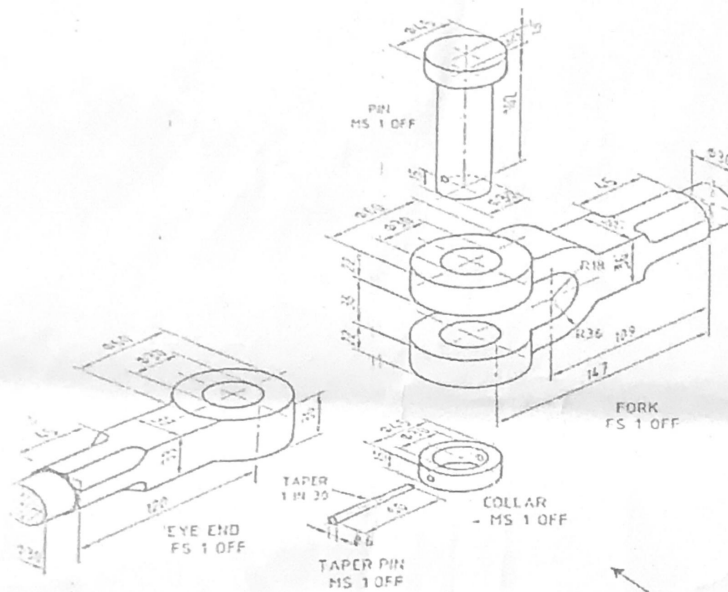


Figure 1: Knuckle Joint

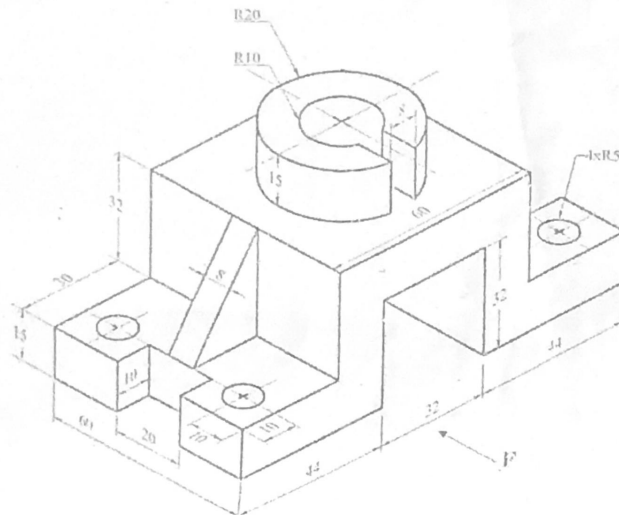
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